Month 4 progress report

Baakhapaa: AI-powered pre-production intelligence system Weeks 13 to 16 · Prepared for Baakhapaa · Reference: `MONTH_4_PROPOSAL.md`

The proposal promised four things. One was engineering, and it is done: **the short-form scripting system now exists as a measured layer inside the product**, not as a separate one. The other three were not engineering, and none of them moved. No unproven path has run on the deployed system, no writer outside this project has finished a script in it, and the proposal's success measure was therefore neither demonstrated nor disproved.

That is the same gap Month 3 closed on, **built against proven**, and this report keeps the two apart all the way through. Everything it says was built was checked in a browser or by a test. Nothing it says was proven was proven.

The building went further than planned: **186 of the 200 tasks in `MONTH_4_TASKS.md` are complete**, and about twenty defects that were on no list were found by walking the product rather than by reading it. Section 4 treats that as the month's second result.

1. What the month produced

Planned building tasks complete	**186 of 200** (`MONTH_4_TASKS.md`, counted by `check_docs.py`)
Proposal deliverables met	**1 of 4**: the short-form layer. The other three need the deployed system, money and writers
Commits on the branch since `codebase`	55
Suite size	**1,319 backend tests in 87 files; 1,585 frontend tests in 80 files**
Defects found that were on no plan	About 20, half of them found only by using the product in a browser
Retrieval, real queries	**91.3%** combined (Month 3). Two craft entries were added and **not re-measured** (section 6)
Real money moved	None
AI generation on the deployed system	None

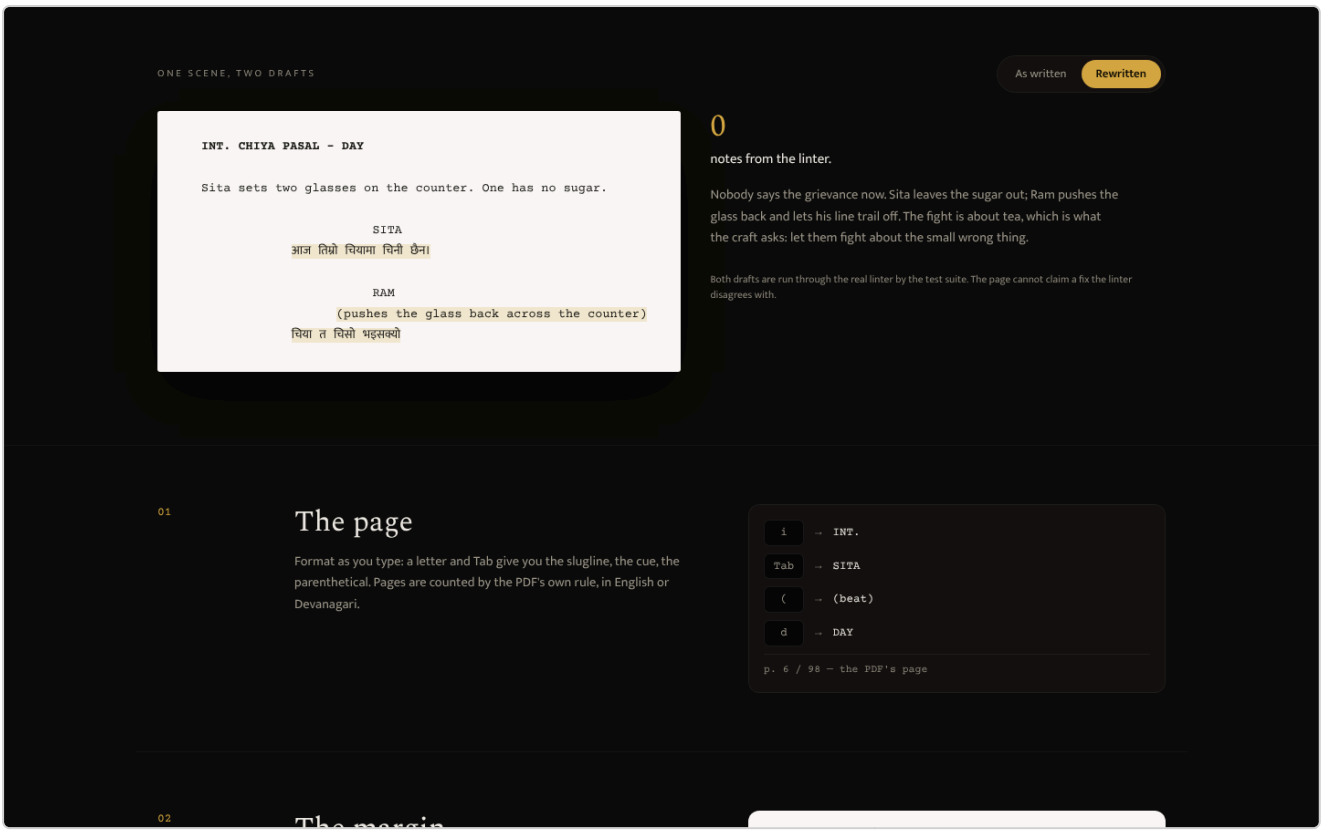
2. The page a stranger sees

`/` used to redirect to `/dashboard`, which bounced a visitor to a password field. A stranger's first sight of the product was a login form.



It is now one claim, "Screenwriting software that reads Nepali", with the product proving it underneath: the editor drawn in HTML, a Devanagari scene typed onto the page, and the linter's notes arriving in the margin beside the lines they name. That is the linter's **real output**, not an illustration: a backend test lints the same scene and fails if the page and the linter ever disagree. The three numbers under it each print the script in the repository they come from, and the three rooms below draw the same screen replicas as the tour, from one shared file. There are no testimonials, because there are none yet, and a test pins that too. Crawlers that run no JavaScript now read 1,564 characters of it instead of 33. First paint on throttled 4G is 884 ms, and that check is now a CI step.

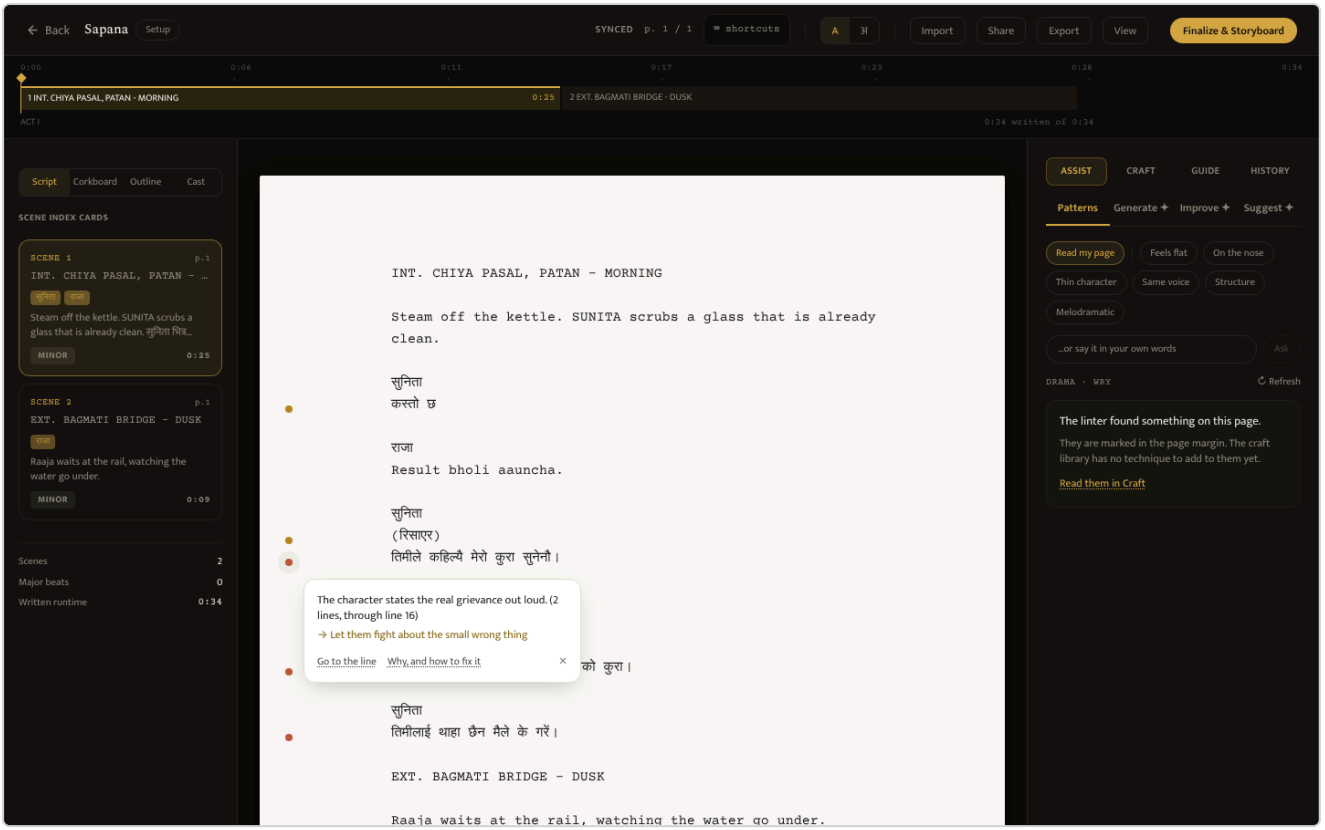
`/tour` is the second public page, for the visitor who wants to see the product work before giving an email address:



One scene in two drafts. "As written" carries the linter's two notes. The rewrite carries none, and **the test suite requires the linter to agree**: the page cannot claim a fix the linter disputes. Other software (Final Draft, Celtx, Arc Studio) is named only as somewhere the exported file opens, with a line saying there is no partnership. Nothing on it says what any other product lacks.

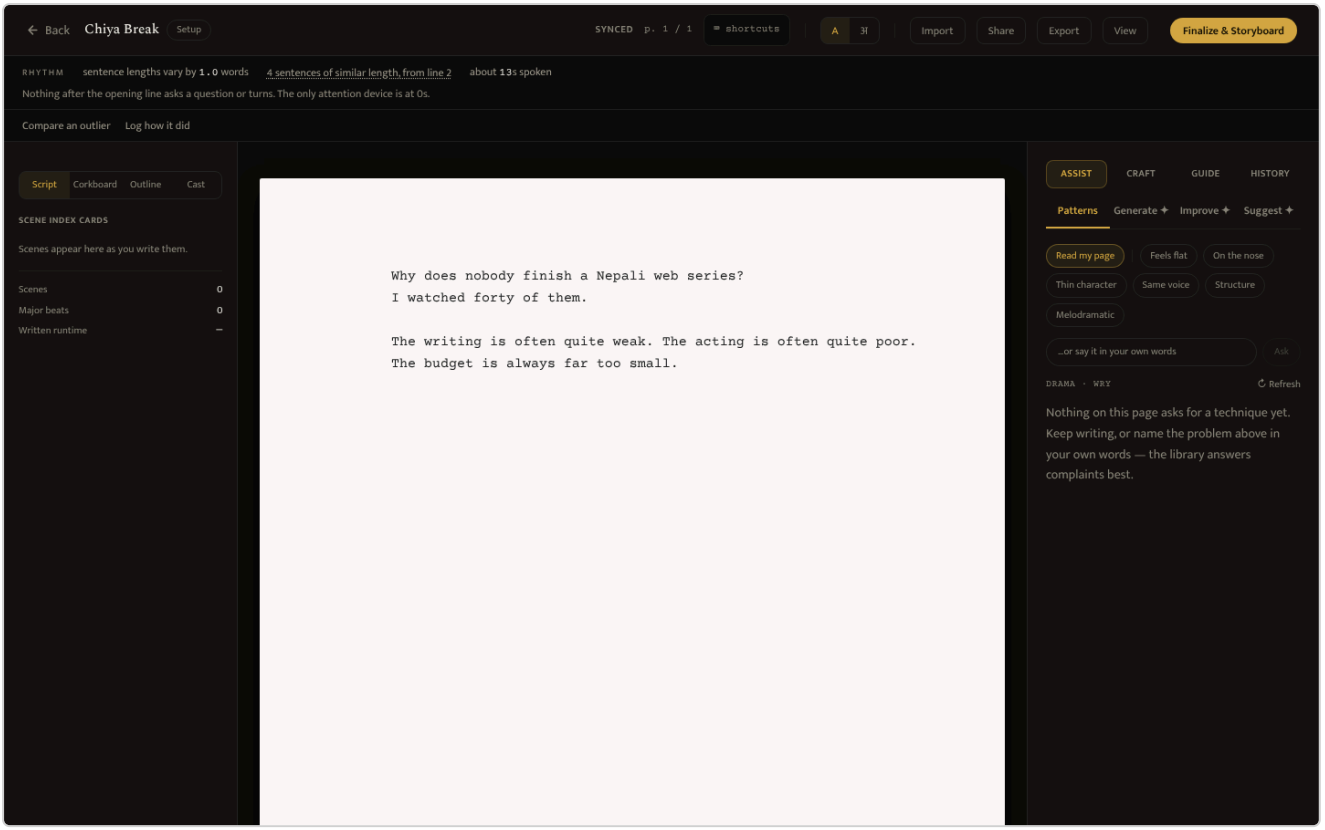
3. What a writer meets on the page

Notes in the margin. The linter's findings lived only in a side tab, as "L13 · The character states the real grievance out loud", so a writer had to open a panel and count down the page. Each flagged line now has a quiet dot beside it, which opens below the line to show what was seen and the technique that answers it.

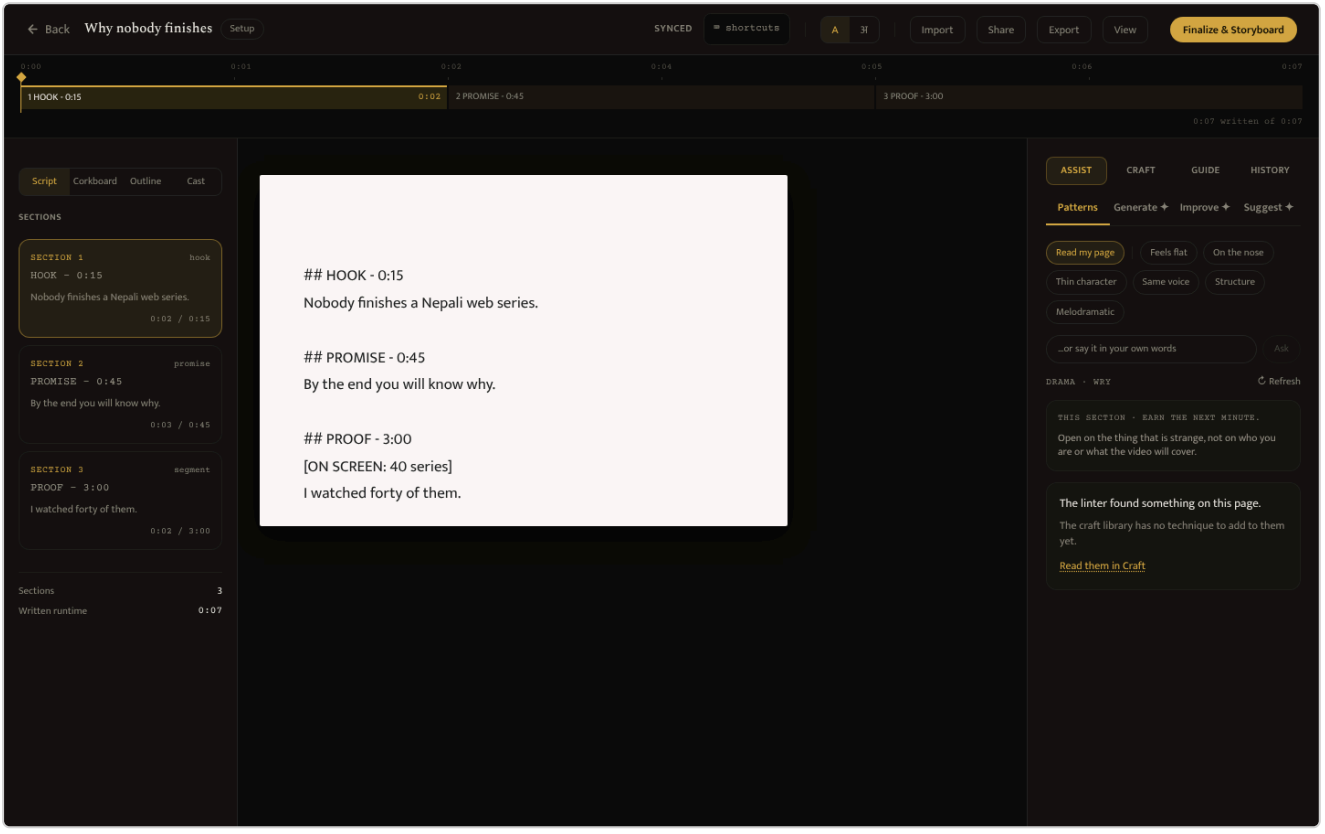


The Assist tab used to say "No patterns matched — try writing a little more first" over a page the linter had just flagged three times. An empty library answer is not an empty diagnosis, and the two now say different things.

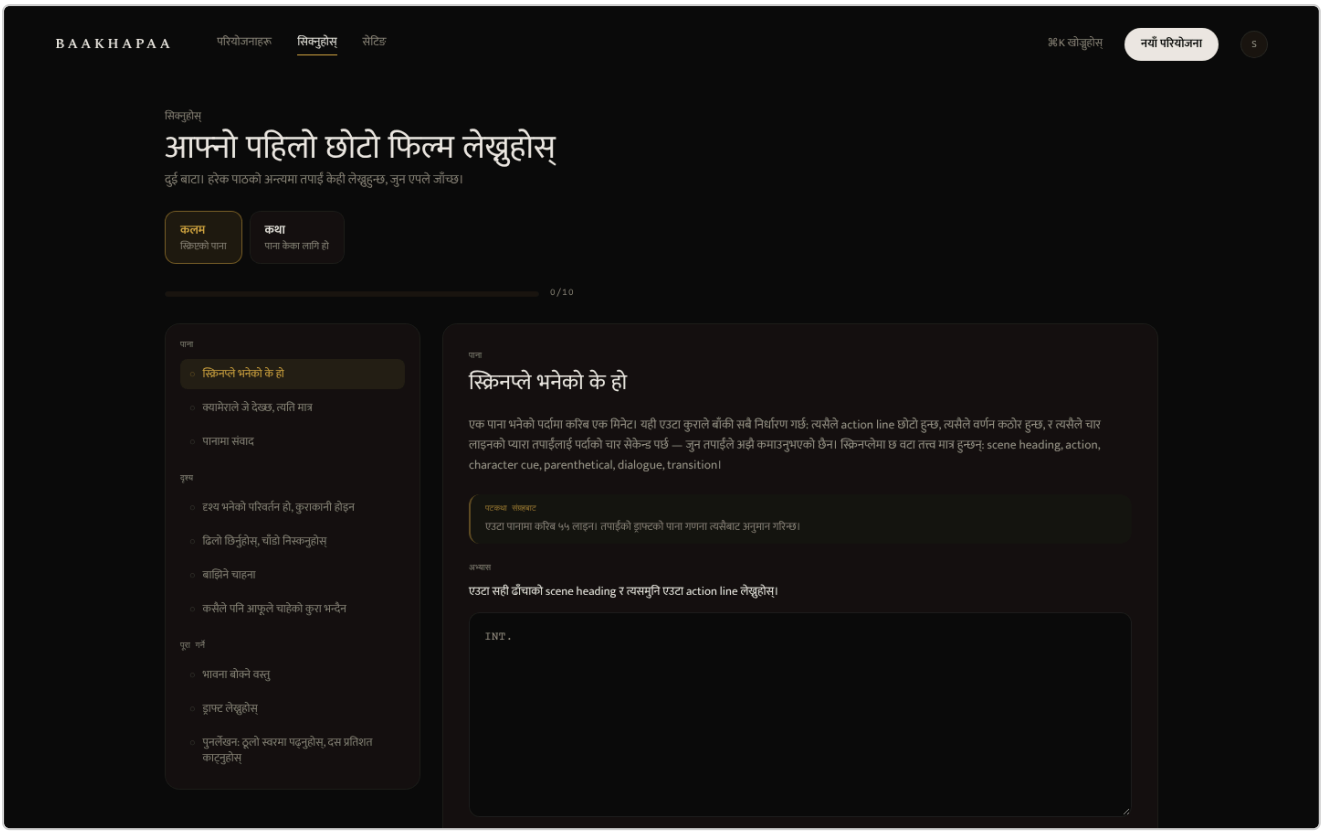
Short-form, measured. The rhythm meter reads the draft as it is typed. It shows the spread of sentence length, names the flattest run by line, and notes when nothing after the opening line asks a question or turns. The beat spine marks where a rehook is due (every gap 5–8 s, on any runtime from 20 s to 3 min), and a creator can paste an outlier's transcript or log how their own post did. Only numbers are stored, and a test reads the stored rows to prove it. Below five results on each side of a comparison, the product says nothing about them at all.



Long-form video stopped borrowing the screenplay. A video section was shown with a page number, an "ACT I" label and screenplay milestones ("3 scenes, and none of them is marked a turning point"). All three were invented for a format that has none of them. A section now shows its kind and its written time against its target.



Nepali, past the prose. A signed-in writer could not change the interface language inside the editor, and the editor toolbar was hard-coded English. The course's module headers were English too. Worst, a writer doing the course in Nepali **passed in Nepali and failed in English**: the grader's explanations were never translated. All nine linter rules and the four grader messages now answer in Nepali, and a test runs the real linter so that a reworded rule fails the test instead of quietly falling back.



4. The defects that were on no list

The plan named four defects on Day 18. About twenty more were found, and the way they were found matters more than the count: **most were invisible to a suite that passed**. They showed up only when the product was opened in a browser, at a phone's width, in Nepali, or through the interface instead of the API.

The ones that would have stopped someone:

- **Course mode was unreachable.** It was built, tested and wired, but nothing in the interface set the video category, so every long-form project was an essay. Found by trying to walk a course lesson through the UI. - **Opening the editor from a project link** loaded the script without its project. That switched off every format-specific feature for the session and titled the script "Untitled". - **A letter typed straight after Tab** landed outside the B-roll cue Tab had just opened, because the caret moved one frame late. - **Devanagari words were split at every vowel sign** by the rhythm and outlier measures, so a five-word sentence counted as twelve, and the danda ended no sentence. - **Storyboard images were the one AI cost the monthly ceiling never counted.** They now spend credits, and the ceiling is shown to writers as credits instead of dollars.

The ones that would have misled:

- The new-project button said "Generate Project Structure" months after the wizard stopped generating one. - The free-plan limit named `/pricing` as plain text and never mentioned the free way out. - Letter-spaced capitals pulled Devanagari conjuncts apart. - The layout check measured the width of the letter "x" in the system's fallback font. It would have failed on Linux CI while passing on Windows. - A test shipped on the branch that could never pass, because jsdom has no layout. Another could never fail: it fed a complete script to a check for an incomplete one.

Every fix has a test that was seen to fail without it. `scripts/walk.mjs`, which found the first group, is kept: it walks five journeys through the interface and records every console error, failed request and stuck step.

5. Testing

	Month 3	Month 4
Backend tests	1,146 in 75 files	**1,319 in 87 files**
Frontend tests	1,281 in 66 files	**1,585 in 80 files**
Browser checks	layout, contrast, responsive, load race	+ landing page (CI), typewriter band, the walk

One backend test does not run in the cloud session: it needs the embedding model, which that session's network policy cannot download. It runs in CI.

6. What is not done

	Waiting on
Retrieval re-measured with the two new craft entries	The embedding model: CI on the open PR, or a machine with it
`short_measurements` table on the real database	Applying the migration in `supabase_schema.sql` in the SQL editor
The two new entries in production retrieval	`load_knowledge_base.py` re-run against production
On-the-nose rule widened from evidence	Real drafts from the pilot (`PILOT.md` now asks for them)
The SDD ledger's deferred items	The ledger is in `.superpowers/`, which only exists on the original machine
AI generation and storyboards, live	Paid API credit
A completed customer payment	Khalti and eSewa merchant accounts
Renewal emails	An SMTP account
The legal documents	A Nepal-qualified lawyer

The first five are small and mechanical. The last four are the same four that closed Month 3, and none of them is engineering.

7. Month 5

Merge, and let CI measure what this session could not. The pull request runs the retrieval eval with both floors. Screenplay p@1 must not move, and the new video entries must hold their own.

Then the proposal's three unmet deliverables, unchanged. Buy the credit, submit the merchant applications, and run the five-writer pilot from `PILOT.md`. Building has gone as far as building can take this. The next number worth having is how many writers finish a script, and only the pilot can produce it.

Figures were captured from the running system with `baakhapaa-frontend/scripts/capture-screenshots.mjs --only m4`, which leaves the earlier months' figures untouched. Test and task counts were derived by `baakhapaa-backend/check_docs.py`, not typed.